

TECHNICAL PROPOSAL

AI-Enabled Modernization of Mystery Shopping Workflow

A phased approach to building a next-generation, purpose-built platform for HS Brands Asia — from mobile data capture to AI-driven quality control.

PREPARED FOR

HS Brands Asia

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Indicative Proposal



Table of Contents

01	Executive Summary	3
02	Current Understanding of the Opportunity	4
03	Proposed Solution Scope	5–6
04	Recommended Delivery Approach	7–9
05	Indicative Investment	10

Working Assumptions

This proposal is based on the following assumptions, which can be refined during discovery:

- HS Brands does **not currently have an existing shopper app** — the platform will be built from the ground up.
- The admin/client-facing process will be primarily **web/dashboard oriented**
- Shopper-side features will be **mobile-first**; admin features will be **web-first**
- Initial rollout will be **English-first** with multilingual support in later phases
- Final integration scope may vary depending on hosting setup and data requirements

Executive Summary

The opportunity is not simply to add isolated AI features, but to build a modern mystery shopping platform from the ground up — so that data collection becomes faster, more accurate, and scalable.

Rationale for Modernization: The goal is to empower clients with comprehensive experience reports based on a single shopper or all configured shoppers per store (including hour-based tracking). Furthermore, this platform will target small business owners by providing dedicated client app access, and deliver a seamless internationalization experience for clients, shoppers, and service providers alike.

Today, the process relies on manual shopper submissions, laptop-based reporting, manual quality checks, and delayed validation. This leads to slower turnaround times, inconsistent data quality, and avoidable operational overhead.

Our recommended approach is to build the platform in phases, starting with the highest-impact parts of the pipeline: the **shopper-side submission experience, validation and QC workflow, admin/client reporting**, and a **scalable cloud-backed architecture** for AI modules.

***Please note: Final timelines will be committed only after the completion of the Discovery Phase.**

Target Objectives

- Submission accuracy & Turnaround speed
- Quality control efficiency & Operational scalability
- Client-facing reporting readiness
- **Client Onboarding** improvements
- **Multi-tenant / Globalization** support

Current Understanding of the Opportunity

Identified Workflow Stages

1. **Setup** — Survey/questionnaire creation, shopper guidelines
2. **Fieldwork** — Shopper identification, assignment, task completion
3. **Quality Control** — Validation, image review, clarifications
4. **Client Reporting** — Dashboarding, insights delivery

Key Pain Points

- **Inconsistent Data Collection:** Manual entry leads to incomplete or contradictory submissions
- **Delayed Quality Control:** Invalid image uploads or poor responses discovered too late
- **Reporting Bottlenecks:** Manual validation processes delay final insights
- **No Existing App:** Lack of a dedicated platform limits operational efficiency
- **Limited Intelligence:** No AI leveraged to extract insights from voice, images, or unstructured data

Proposed Solution Scope

To address these challenges, YOM Solutions proposes building a proprietary end-to-end platform structured around four core pillars:

The Shopper Application (Mobile-First)

01

A purpose-built mobile app designed to reduce friction and improve data accuracy in the field.

- **Location-Aware Tasking** — Intelligent, geo-fenced assignment suggestions based on shopper proximity
- **Voice-to-Text Reporting** — Frictionless post-visit surveys powered by conversational AI
- **Smart Submission Engine** — Real-time, on-device validation catches missing fields and contradictions before final submission
- **Automated Media Checks** — On-the-fly verification of image quality and basic relevance
- **Guided Submission Flow** — Structured prompt-based data capture to reduce ambiguity
- **Geo-Verification** — Link tasks to store location and geo constraints

HS features — Task Approvers, QC, Validation

02

Built to supercharge the QA team and streamline program management.

- **AI-Assisted Quality Control** — Auto-flagging of suspicious or incomplete submissions
- **Smart Review Queues** — Intelligent prioritization based on urgency and risk scores
- **Operational Command Center** — Real-time visibility into submission statuses, validation pipelines, and shopper performance
- **Report-Readiness Workflow** — Faster determination of when submissions are client-ready

The Cloud AI Foundation

03

The secure, scalable backend powering the entire ecosystem.

- **Centralized Data Lake** — Secure storage for images, audio, and structured report data
- **Scalable Processing** — Cloud infrastructure designed to handle peak reporting periods without degradation
- **Backend APIs** — Unified APIs for mobile, admin, and AI modules
- **Enterprise-Grade Security** — Role-based access controls, audit trails, and data privacy compliance
- **AI Pipeline Architecture** — Modular AI services for voice-to-text, image validation, and intelligent routing

The Client & Admin Portal

04

Faster, more reliable access to validated submissions and flexible onboarding.

- **Cleaner Submissions** — Structured data arriving seamlessly from the shopper app
- **Reduced Manual QC** — Less workload through AI-assisted flagging and validation
- **Real-Time Visibility** — Operational dashboard enhancements for immediate status tracking
- **Client Onboarding** — Faster setup and dedicated tenant management for small businesses
- **Report Readiness** — Improved downstream reporting and faster client handoff

Recommended Delivery Approach

We recommend a phased implementation model. Each phase delivers measurable value while keeping risk manageable.

Phase 1 — Discovery & Technical Blueprint

Objective: Understand the current operation, validate assumptions, and define the implementation blueprint.

- Schedule calls with stakeholders and business to understand current process
- Document As-Is flows and process maps
- Define technical approach for AI integration and platform architecture
- Review privacy, governance, and data flow considerations

Deliverables: Discovery summary, As-Is flows, technical blueprint · **Timeline:** 2 weeks

Phase 2 — Shopper App & Submission Modernization

Objective: Build the shopper mobile application and modernize how shoppers complete and submit visits.

- Cross-platform mobile app development
- Voice-based post-visit survey flow with AI transcription
- Guided questionnaire UX, image upload workflow, geo-linked task logic
- Submission-time validation engine

Deliverables: Shopper mobile app, voice-enabled survey, smart submission flow · **Timeline:** 13 weeks

Phase 3 — HS Process Improvements

Objective: Build the admin/validator dashboard and reduce manual effort through AI-assisted quality control.

- Image validation checks, auto-flagging of incomplete submissions
- Review queue improvements, validation status indicators
- Admin-side submission visibility and report-readiness workflow

Deliverables: Admin dashboard, partial QC automation, faster validation · **Timeline:** 8 weeks

Phase 4 — Client Onboarding & Admin Solutions

Objective: Empower clients and small business owners with dedicated access, robust configuration, and deeper insights.

- Client onboarding changes and tenant provisioning
- Web and App support dedicated for clients
- Enhanced configurable reports for clients
- Manage storefronts functionality
- Customize questions and dynamic survey building for clients

Deliverables: Client app/web portal, configurable reporting, onboarding flow · **Timeline:** 5 weeks

Phase 5 — Platform, Cloud & Intelligence

Objective: Strengthen backend scalability and add operational intelligence features.

- Backend API enhancements, cloud media/data storage setup
- Role and access management, logging/audit setup
- Location-based task suggestion logic, shopper matching rules
- Groundwork for internationalization and multilingual workflows

Deliverables: Scalable backend, location-based task suggestions, AI pipeline foundation ·

Timeline: 4 weeks

Phase 6 — Testing, Pilot & Stabilization

Objective: Validate new workflows in a controlled rollout and refine based on real usage.

- Quality assurance and user acceptance testing
- Pilot rollout with select shoppers and clients
- Bug fixing, hardening, and feedback-driven iteration

Deliverables: Pilot-ready release, stabilized production workflow, rollout recommendations ·

Timeline: 3 weeks

Total estimated timeline: ~35 weeks (~8 months)

Phases are sequential. Some overlap is possible between phases to accelerate delivery.

Indicative Investment

DEVELOPMENT PHASE	FOCUS AREA	ESTIMATED COST (USD)
Phase 1 — Discovery & Blueprint	Architecture, As-Is flows, Planning	\$6,500
Phase 2 — Shopper App & Backend	Mobile app, submission, databases	\$46,000
Phase 3 — HS Process Improvements	Dashboard, AI validation engines	\$26,000
Phase 4 — Client Onboarding	Tenant provisioning, reporting, web/app	\$17,000
Phase 5 — Platform & Intelligence	Routing, global groundwork, APIs	\$14,000
Phase 6 — Pilot & Stabilization	UAT, bug-fixing, deployment	\$10,500
Total Estimated Investment		\$120,000

Final investment depends on: integration complexity, level of AI model tuning required, data volume, and hosting requirements confirmed during discovery.

We believe the strongest initial value lies in building the **shopper mobile app**, combined with **submission-time validation** and **QC automation**.



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